



OPEN NAVIGATION

**Robotics Workload Platform Benchmarking
Results**

v1.3

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Introduction

This report details the robotics workload benchmarking results to compare the AMD Strix Halo (X199) with NVIDIA Jetson Thor 5000 and Orin AGX platforms. This benchmark was designed to test each platform using realistic computational loads of a fully-autonomous robotics product in a medium-scale warehouse environment to benchmark the performance of a representative deployed application. This better displays the fully-loaded performance, shortcomings, and characteristics of each compute platform as would be seen by a scaled, end-use customer. This does not replace isolated microbenchmarks of computational power and efficiency, but augments it to provide an additional axis of data for the selection process.

The design of the benchmark and results are detailed in the sections below. In summary, Strix Halo performs generally on-par with GPU performance to the NVIDIA Jetson Thor but many times better in CPU performance using the powerful x86 cores, leaving substantial room for additional loads, application programs, or increasing data throughput for higher performance robotics applications.

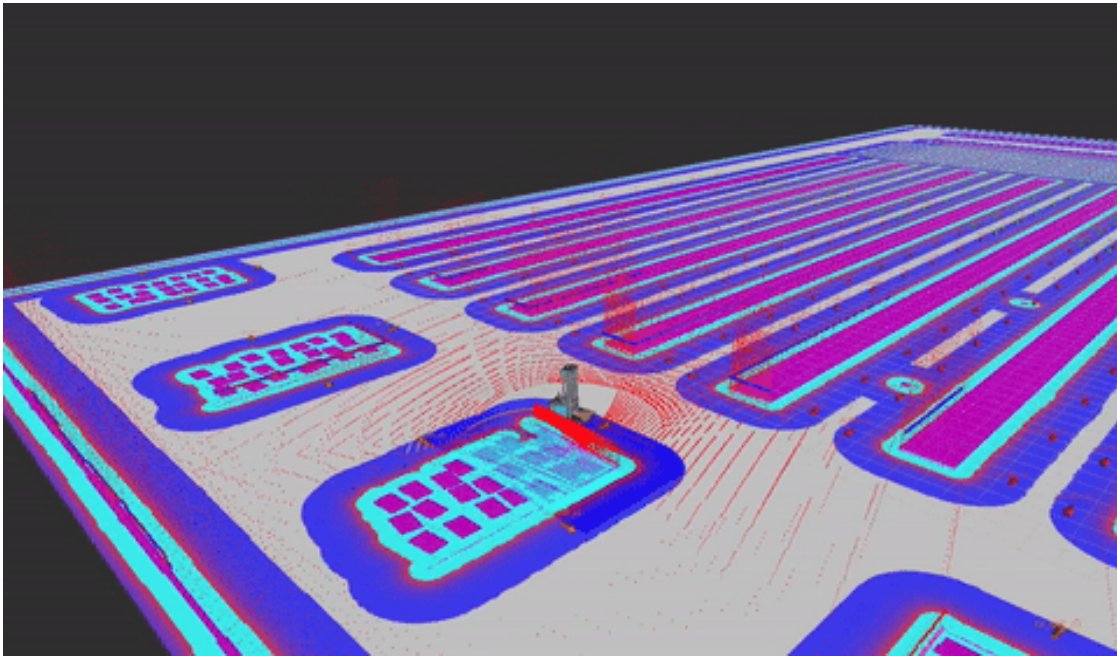


Figure 1: Robot, Data in Simulated Environment

Click on image to see a full video

Benchmark Design

This benchmark is designed to represent a realistically complex robotics system, such as those currently deployed around the world. This experiment uses a material handling forklift configured to move at 2 m/s outfitted with a broad range of modern sensors, including:

- 2x 2D Safety LIDAR (30 Hz, 25m range)
- 3x 3D LIDAR (10 Hz, 30m range, 32 beams @ 512 resolution)
- 3x RGBD Cameras (10 Hz, 320x240 resolution)
- 1x RGB Camera (5 Hz, 640x480 resolution)
- 1x IMU (100 Hz)
- 1x Differential Drive Controller

This sensing stack provides a significant amount of data that needs to be processed by navigation and application software stacks to safely navigate complete important mission tasks. The 3D LIDARs in particular are more commonly being deployed as the cost of these sensing platforms are dropping, yet many basic simulations or previous benchmarking neglects them. RGBD cameras are commonly used for localized collision avoidance at low ranges, with a high resolution relative to other sensors. The 2D LIDARs are used to simulate safety sensing while the IMU is used for inertial measurements to provide accurate state estimation. Finally, the RGB camera is used as the input for VLM or another AI models, located in a central, forward location for high visibility.

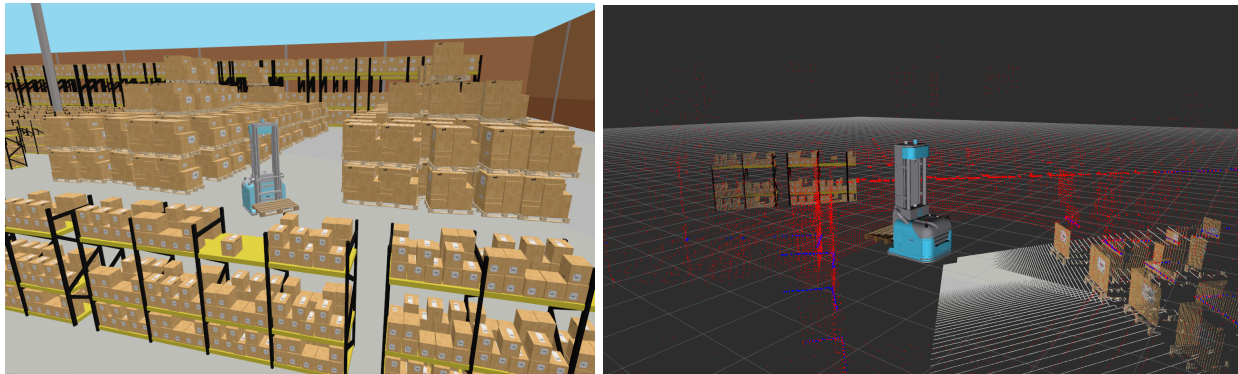


Figure 2: Material Handling Forklift Robot in Simulation (Left) and Live Data Produced (Right)

The workload consists of two major components: a *robotics workload* and an *AI workload*.

The robotics workload is created to mimic a pick-and-place task for a forklift robot to move pallets from shipping & receiving areas to designated shelving units for storage. The robot is assigned random missions from one of several pick locations in block-stack style pallet configurations in a loading dock region into large racking. Every 100 paired missions, the robot returns to the charging dock for 30 seconds to simulate recharging the batteries before continuing to the next mission.

This workload conducts all of the major elements of a robotics product:

- Localization in the environment using 2D LIDARs
- Perception processing all 3x 3D LIDARs, 2x 2D LIDARs, 3x RGBD cameras, and IMU
- Path Planning across a 200,000 sqft indoor warehouse facility
- Optimal Control using Model Predictive Path Integral (MPPI) Trajectory Planning
- Autonomous navigation behavior using behavior trees leveraging AI results
- Fault and error handling to ensure reliable mission completion
- Collision monitoring and velocity smoothing for high-quality repeatable behavior

The AI workload consists of the Gemma 4.0 31B parameter VLM model Q4_K_M quantization used on the RGB image streams to discern key information about the environment. This is used for scene understanding and regional context to adjust the robot's behavior based off of. Queries to the VLM regarding the state of the environment (such as crowds, blockages, spills, or other hazards) are used in the autonomous navigation behavior tree to adjust the robot's path, speed, control algorithm, and more. While the behavior tree can be quite reactive based on the VLM responses, these updates are relatively infrequent due to the runtime the VLM model takes (20-30s) on these Edge compute platforms. VLM queries are immediately requested after the previous query completion, effectively saturating the GPU resources of the platforms being tested with a continuous AI workload.

The robotics and AI workloads are each contained in Dockerfiles compiled on each platform for use to ensure consistent benchmarking and launched from a `run_benchmark.sh` script. The simulation is instead run on a connected computer to produce sensor data and act as the physical robot to completely isolate the simulation load from the platform under test, who is acting as the robot's 'brain'.

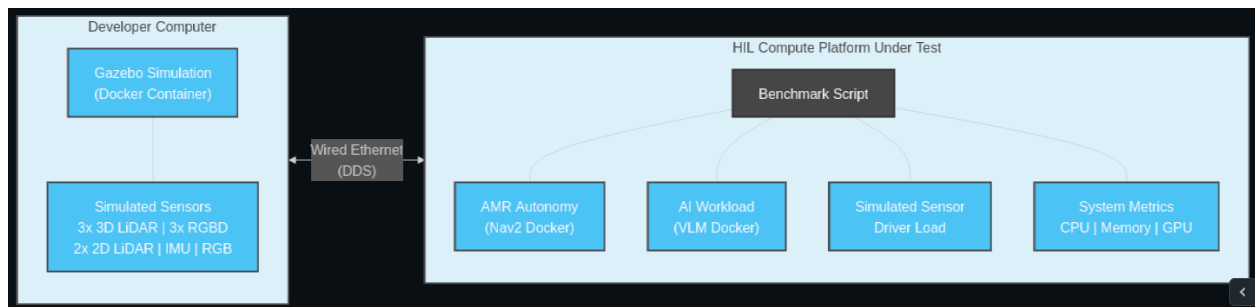


Figure 3: benchmark Experimental Design

An additional workload to simulate sensor driver loads are also added to each platform to ensure the effects of these are represented in the benchmark. Typically, RGBD camera and 3D LIDAR drivers are non-trivial loads on the system and the benchmark would be incomplete without simulating these loads. The loads of these were measured using real world hardware (Orbecc Gemini 355 and Ouster OS-1 32) on each platform. Then, a load is added to each platform under test to represent the number of sensors and each of their respective loads.

Finally, programs are run in the background of a platform being tested to measure system metrics for later analysis.

Analysis

We run the benchmark in two power settings on each platform to obtain key results. The max power TDP is used to showcase the upper limits of what is possible on each platform while the balanced power TDP is used to showcase an apples-to-apples comparison of each at as similar of a power level as possible. The full data and reports can be found in the the [GitHub repository](#).

Max Power TDP

The max power TDP for the platforms are as follows:

Platform	Power (W)
AMD Strix Halo	120W
NVIDIA Jetson Thor 5000	130W
NVIDIA Jetson Orin AGX	65W

Note that the Orin is substantially weaker as an older generation of compute platforms. We compare it here to display complete benchmarking as this is a commonly used platform in production applications today.

We can see from Figure 4 below an overview of the results comparing the three platforms. It demonstrates that the Strix Halo (X100) has similar GPU and memory capabilities as the NVIDIA Jetson Thor. While the AMD platform appears to be slightly more powerful in GPU capabilities, these metrics are sufficiently close that from a practical perspective they could be considered equivalently powered. However, Strix Halo demonstrates 2.7x the performance based on available CPU headroom by core (Figure 7) and approximately 400% by available compute headroom per watt as either the Thor or Orin AGX providing substantial headroom to perform additional computations needed for a robotics application, additional sensor processing, or other necessary workloads. Note that Strix Halo using the continuous GPU saturating VLM workload is power-constrained, so to capture that additional CPU resource, some GPU workload may need to be reduced. However it is measured that at least an additional 8 cores can be used with no change.

The Orin AGX is lacking in essentially every metric and provides no usable headroom in CPU for additional applications. In fact, in future charts we will show that the Orin AGX is broadly unable to keep up with the robotics and AI workloads sufficiently well for this application.

We can also see that the Orin is also the least power efficient of the three platforms under test. Strix Halo is the substantially most efficient in total available compute utilizing its x86 cores, but the Thor is 17.5% more efficient in GPU-only compute.

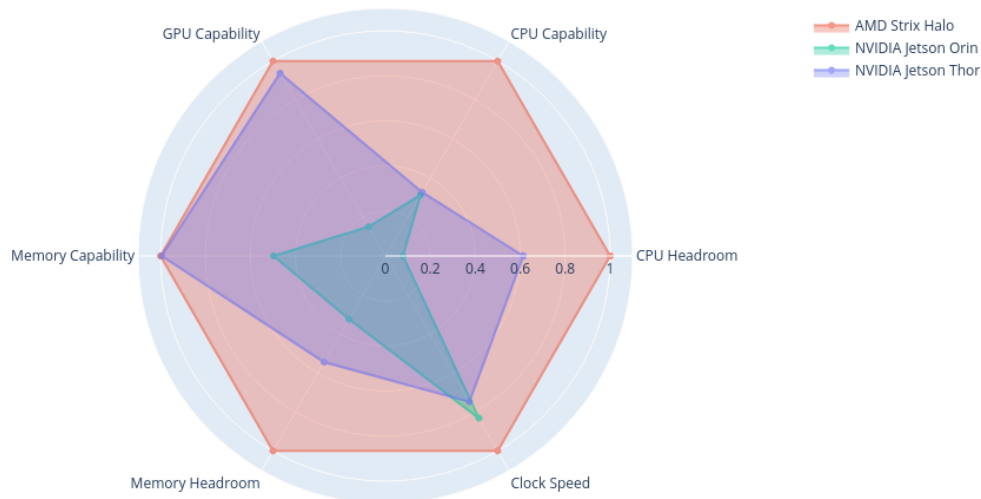


Figure 4: Radar Diagram of Thor, Orin AGX, and Strix Halo Results

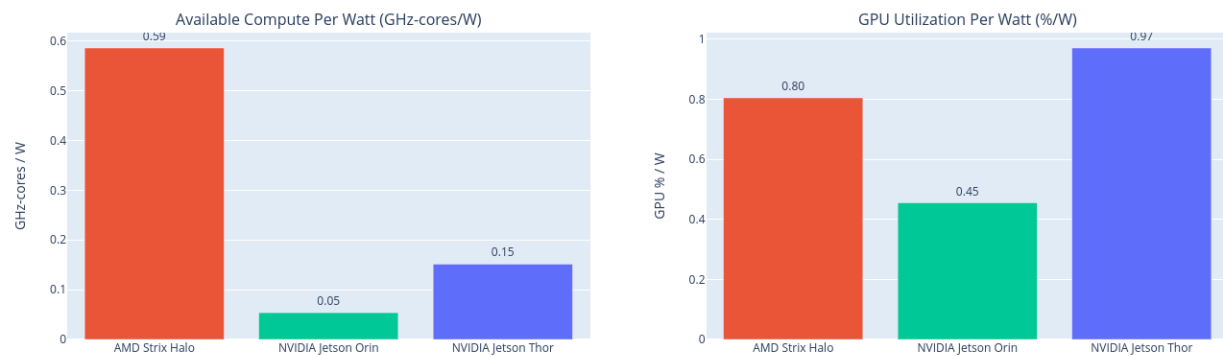


Figure 5: Power Efficiency, Performance Per Watt

The GPU for all 3 platforms are entirely saturated by the VLM workload. Gemma 4.0 31B is the largest dense model provided by Google and easily utilizes the full GPU capabilities. Thus, the GPU headroom or utilization metrics are largely not representative. Instead, we will review the number of completed VLM queries during the benchmark duration. This demonstrates the throughput possible for each platform with the fully saturated GPU.

Below in Figure 6, we can see that the Orin AGX is unable to complete a single query during the benchmark. On inspection of logs, the Orin AGX takes too long to respond to queries that by the time a response is available, the navigation mission across the entire warehouse space is already completed; pre-empting the VLM request. AMD's Strix halo is able to process 47 requests, with 34 completed successfully. Jetson Thor processes 35 total requests with only 25 successfully completed, indicating that Strix Halo has a 26% increase in performance in this experiment. However we note that Thor consumes on average 91W compared to AMD's 120W during the experiment, so the performance-per-watt of both platforms is nearly equivalent (< 5%). We see a similar result play out as well in independent VLM benchmark testing whereas Strix Halo consistently performs VLM requests on robotic scenes in a warehouse environment similar to the Jetson Thor looking at token/second.

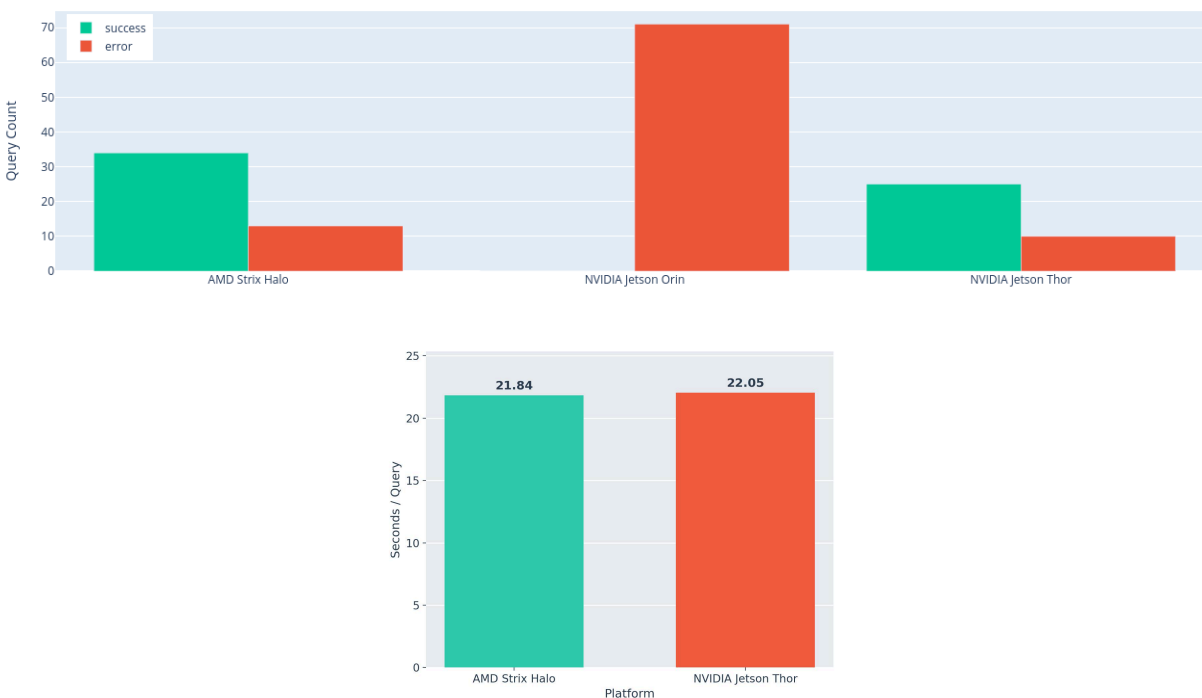


Figure 6: GPU Throughput (VLM Queries)

The CPU performance is captured in the following charts on Figure 7. We can see that the Jetson Orin AGX's CPU, as well as GPU as previously mentioned, is completely saturated at 93.6% mean utilization. This is too high for a real-time application and thus is not able to keep up with the demands of this representative benchmark application due to the VLM and Robotics workloads.

The application showcases this behavior. We can see that the Jetson Orin AGX is unable to complete 7/10 missions given in the 15 minute benchmarking window, while both other platforms are able to reliably complete all missions provided (Figure 8). This is due to the Orin's complete saturation of resources unable to keep up with the demands.

We can measure this more directly in Figure 9 below, where the Orin misses on average 5.8 control loop scheduled times **per second** of the benchmark. The Thor misses 1.6 per second on average, loaning to maxing out CPU cores allocated to the trajectory planning processes with its ARM CPU (Figure 10). While it has sufficient compute bandwidth to take on additional tasks, the cores themselves are not powerful enough to keep up with highly computational operations which cannot be offloaded to the GPU like model predictive control. Strix Halo misses 0.45 per second. Note that Thor and Strix Halo may be improved by setting thread priorities for the control processes.

Thusly both the Thor and Strix Halo however contain suitable, production ready headroom and remaining absolute compute available to handle spikes during run-time without missing key deadlines. However, Strix Halo has notably more headroom which can be used to deploy more advanced algorithms, process more sensor data, move the robot at higher speeds, or otherwise leverage for application-specific tasks - especially when using workloads that do not fully saturate the GPU. Nearly 82% of the CPU bandwidth is still available for application developers representing multiples on the total compute available on either Jetson platform.

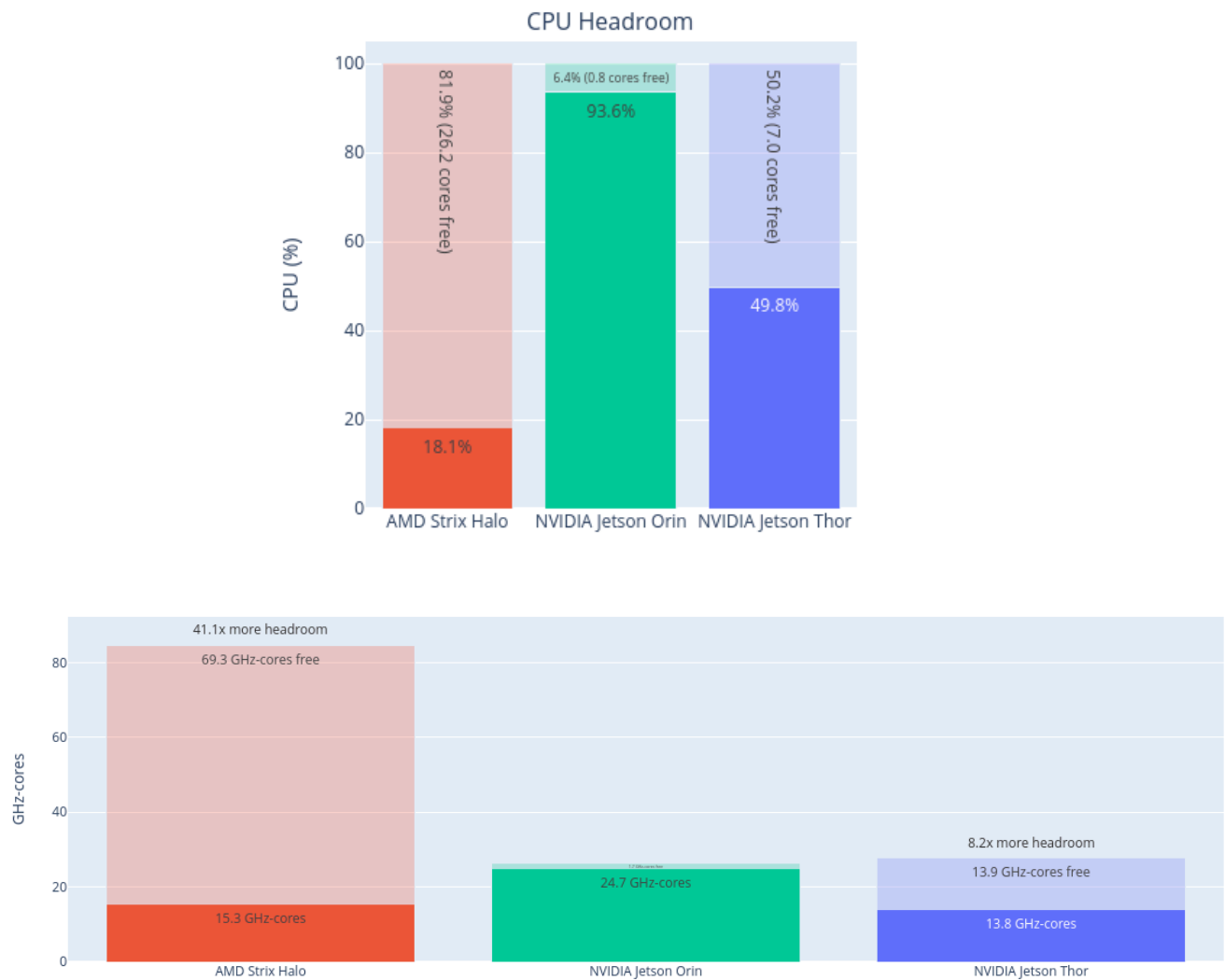


Figure 7: CPU Resource Headroom, % and Absolute

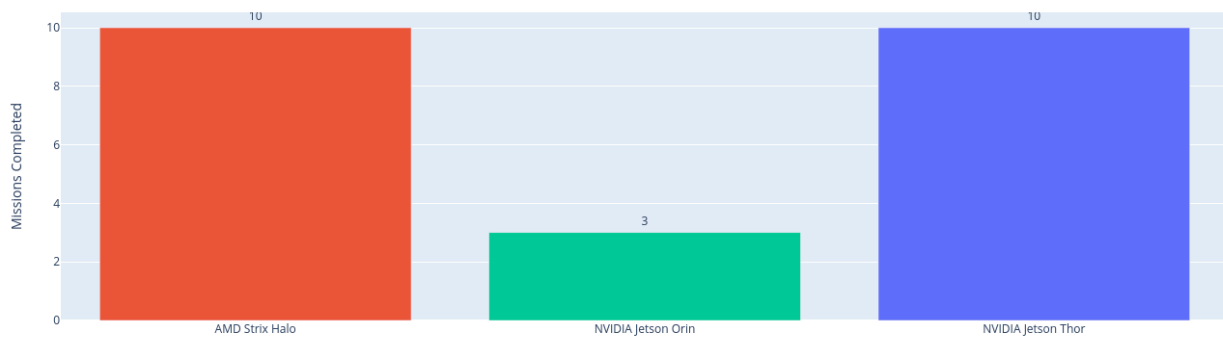


Figure 8: Mission Completions

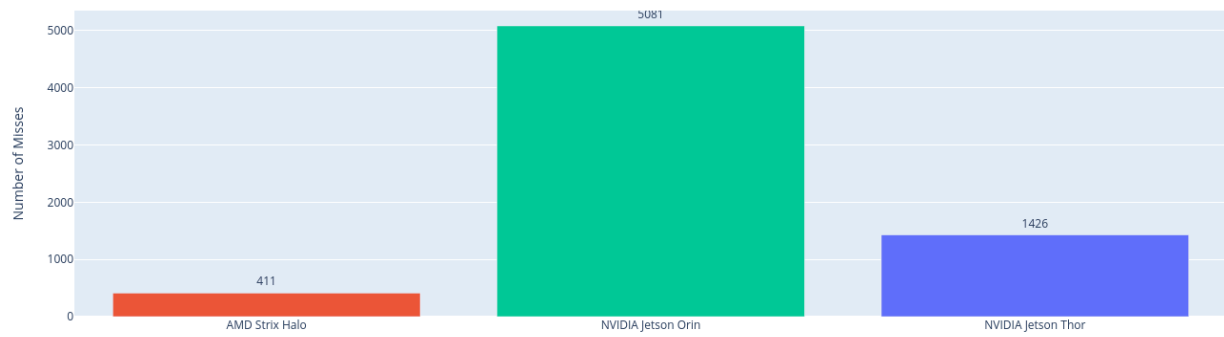


Figure 9: Control Loop Scheduling Misses

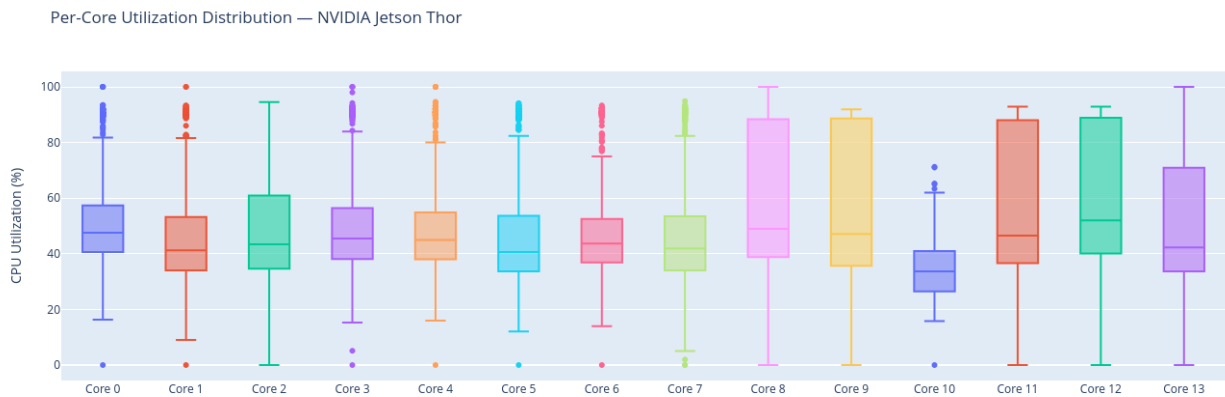


Figure 10: Thor's CPU Core Utilization, Note Core 8, 9, 11, 12, 13 are highly loaded

Balanced Power TDP

The balanced power TDP for the platforms are as follows:

Platform	Power (W)
AMD Strix Halo	85W
NVIDIA Jetson Thor 5000	90W
NVIDIA Jetson Orin AGX	65W

Note that the Orin remains in its max power configuration since it is still weaker.

Figure 11 shows the balanced power TDP comparison for the platforms. We can see consistently with the max power mode that the GPU, memory capabilities of both Thor and Strix Halo are aligned. The CPU capabilities are also broadly higher, but the gap has closed to only 300% more powerful in performance-per-watt and 2.6x in core headroom. The power efficiency per watt also displays the substantial total compute performance efficiency for AMD and closes the gap with respect to max power TDP to only 5.9% on GPU-only efficiency between the Thor and Strix halo. Orin AGX continues to be a distant last place in all power efficiency metrics.

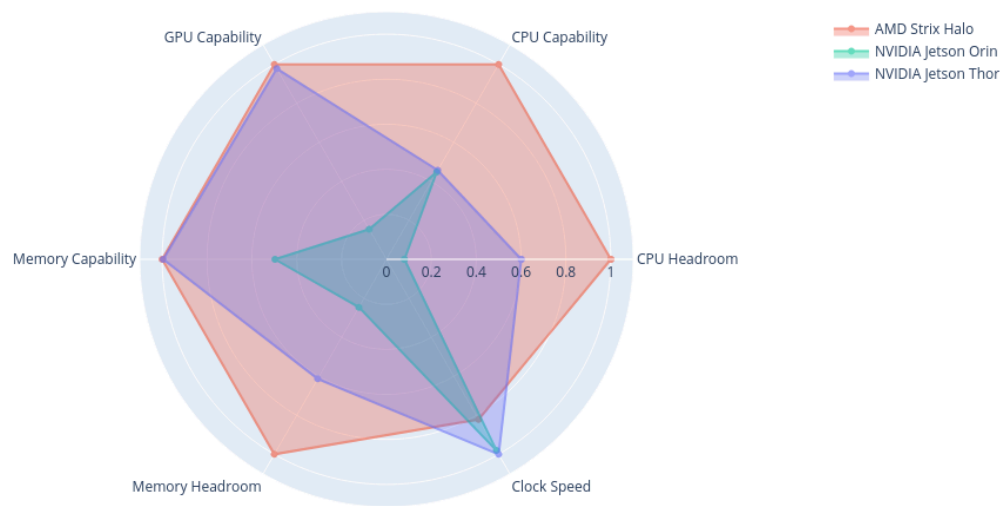


Figure 11: Radar Diagram of Thor, Orin AGX, and Strix Halo Results

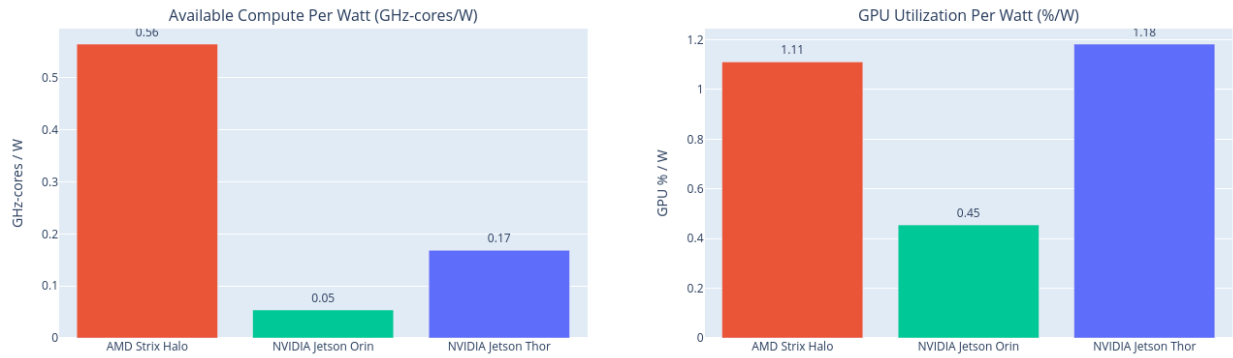
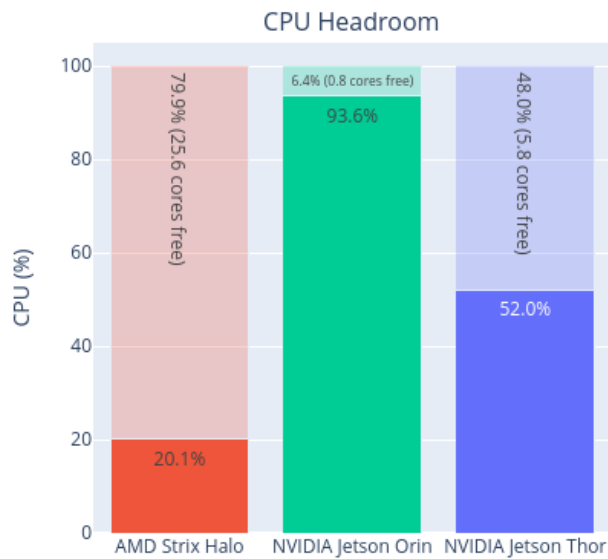


Figure 12: Performance Per Watt, Balanced Mode

This can be seen in more detail below with the CPU percentage and absolute headroom remaining. Note as before the GPU is fully saturated by the VLM task. In the balanced power mode, the Thor and AMD GPU's performance is practically equivalent with no discernible improvement in VLM throughput between them.



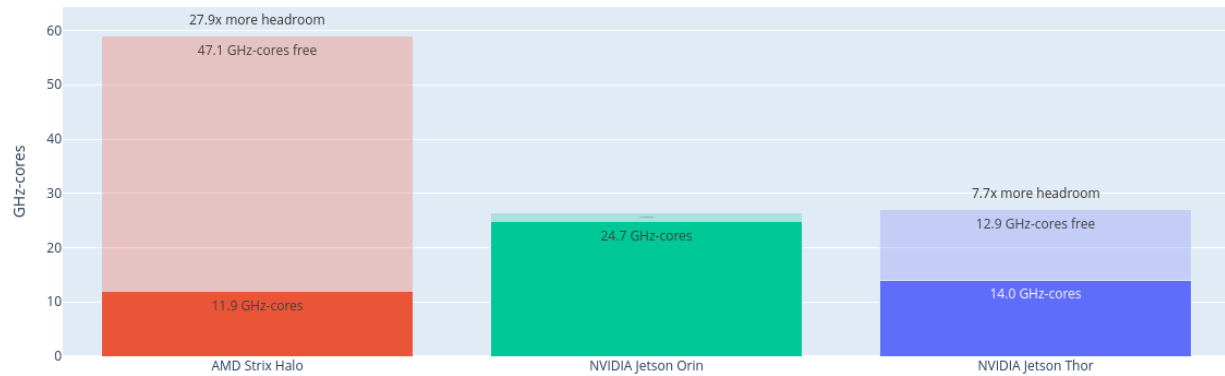


Figure 13: CPU Resource Headroom, % and Absolute

Many of the trends remain the same in the balanced power TDP mode as in the max power mode. Where they differ is in the number of control loop misses. In this case, we see the Strix Halo actually has a larger number of control loop misses for a similar reason as the Thor in the max power mode. The CPU cores allocated to the control loop are saturated and while there is a broad amount of CPU capabilities remaining, the individual core responsible for the control loop was unable to keep up due to the power allocation to the saturated GPU (figure 15). However unlike the max power mode which uses all power available, this on the AMD platform likely could be resolved with fine tuning of the TDP mode settings rather than using the generic 'balanced mode' provided in the BIOS or reducing the GPU workload some.

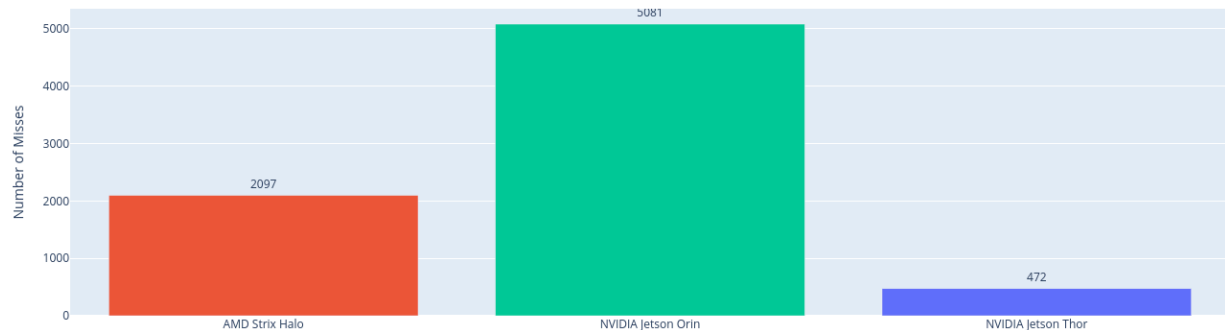


Figure 14: Control Loop Deadline Misses

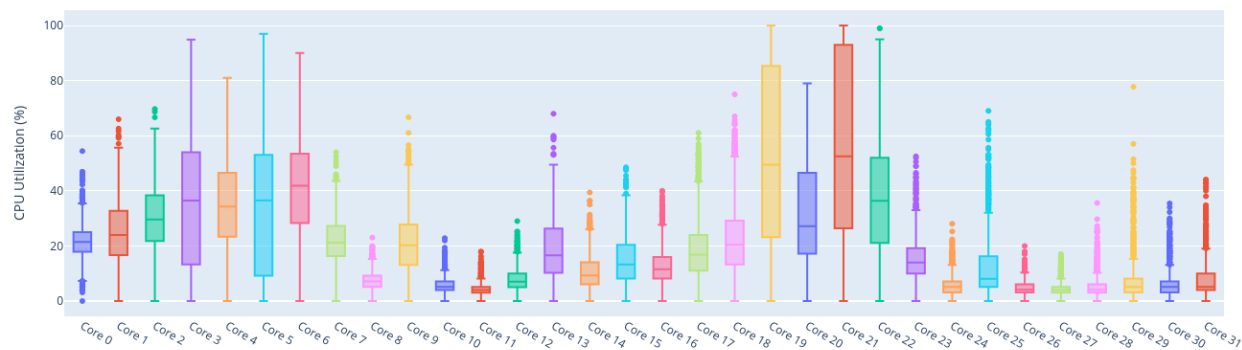


Figure 15: CPU Utilization Per Core on Strix Halo Balanced Mode, note Cores 19, 21

Final Conclusions

In analysis of these platforms, it is abundantly clear that the Jetson Orin AGX cannot keep up with the workloads in any meaningful fashion. It is unable to meet key control rate deadlines, complete missions altogether, and has its complete CPU and GPU resource saturated. The other modern platforms available (like the others tested) are proven to be more powerful and better suited for the needs of contemporary robotics applications - including more traditional industrial automation, warehouse, and factory solutions.

The NVIDIA Jetson Thor and AMD Strix Halo are both powerful and capable modern platforms which can be used to power robotics applications today. In both maximum power and power limited applications, we see that these two platforms can keep up with a demanding GPU application. At either power level, the AMD platform provides substantially more CPU capability for similar power levels which is a key advantage for robotics workloads.

This extra CPU headroom can be used to process more data, faster to enable higher rates of operation and the deployment of the most advanced, highly computational planning and perception algorithms. Since this platform contains best-in-class GPU performance with high performance x86 cores, the AMD Strix Halo is a strong replacement option for mixed applications leveraging multiple less powerful computers (i.e. Jetson Orin + x86) in a single platform reducing product power consumption, complexity, networking, and engineering overhead.

If additional CPU capabilities or stronger CPU cores are not required, the NVIDIA Jetson Thor is a powerful modern option. However it struggles to complete real-time optimization-based trajectory planning algorithms within key deadline constraints in any power mode which could be a limiting factor.

